

gdTAI v3 TRDC/NK Guard Decision Report

Decision: do not promote this candidate as the default gdTAI v3.0 model yet.

The selected candidate meets the requested atlas-level recall and estimated false-positive target, but it fails the explicit NK false-positive promotion gate on the independent external test.

DECISION

Do not promote

The v3 candidate remains packaged but should not replace v2 high-purity yet.

EXTERNAL F1

0.902

v2 high-purity: 0.902

EXTERNAL RECALL

85.9%

above 0.8 target, slightly below v2 high-purity

FULL-ATLAS RECALL

80.56%

primary-gold labels only

ESTIMATED ATLAS FP/PRED

4.06%

below 5% target

EXTERNAL NK FP

9 vs 3

failed promotion gate

1. What changed in this iteration

GSE144469 moved into train/tune

Other validation cohorts retained

NK hard negatives restricted to TRDC+/TRDV-

Independent external H5AD held out

Full 5.13M-cell atlas applied

GSE144469 contributed 63,764 train/tune cells (4,003 gdT positives and 59,761 abT negatives). The non-GSE144469 validation block remained held out with 186,195 cells.

Hard NK negatives follow the requested rule: 1,453 NK TRDC+TRDV- cells from 23 datasets were retained (888 train, 565 tune). NK cells outside this expression rule were excluded from NK hard negatives (31,117 cells). NK+TCRAB-overlap cells were not used as explicit NK hard negatives; 4 expression-passing weak-TRDC overlap cells were also removed.

Step conclusion: the split design now matches the requested policy, and the external H5AD remains an independent final test.

2. What the algorithm is

The selected candidate is `v3_round14_v2_score_trdc_gate_fixed_0p936` at threshold `0.936`. It is **not XGBoost**. It is a conditional guard around the gdTAI v2 transcriptome score, with engineered features for TCR gene evidence, CD3 strength, NK-marker strength, and TRDC-only risk.

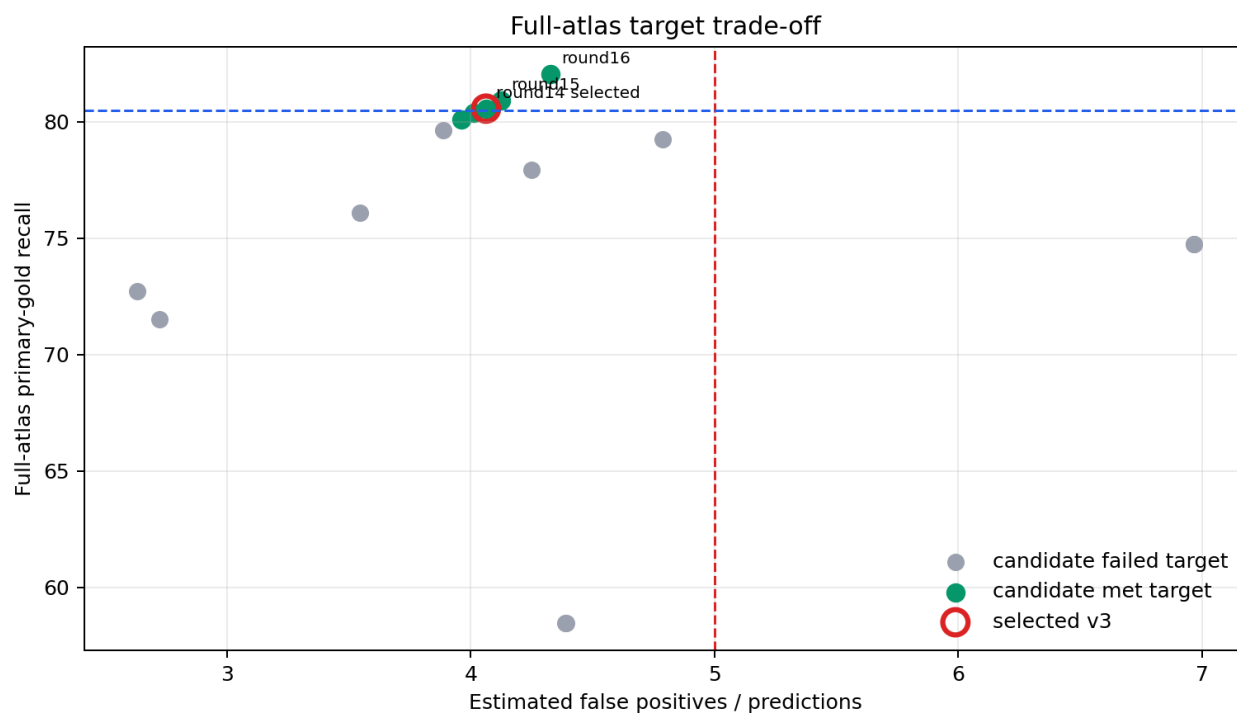
External inference requires raw counts from `layers["counts"]` and computes `log1p(counts per 10,000)`. It does not require TCR-seq metadata at inference time.

Step conclusion: this model is a conservative v2-score-plus-guard candidate, designed to reduce specific false-positive modes without making NK markers automatic exclusion genes.

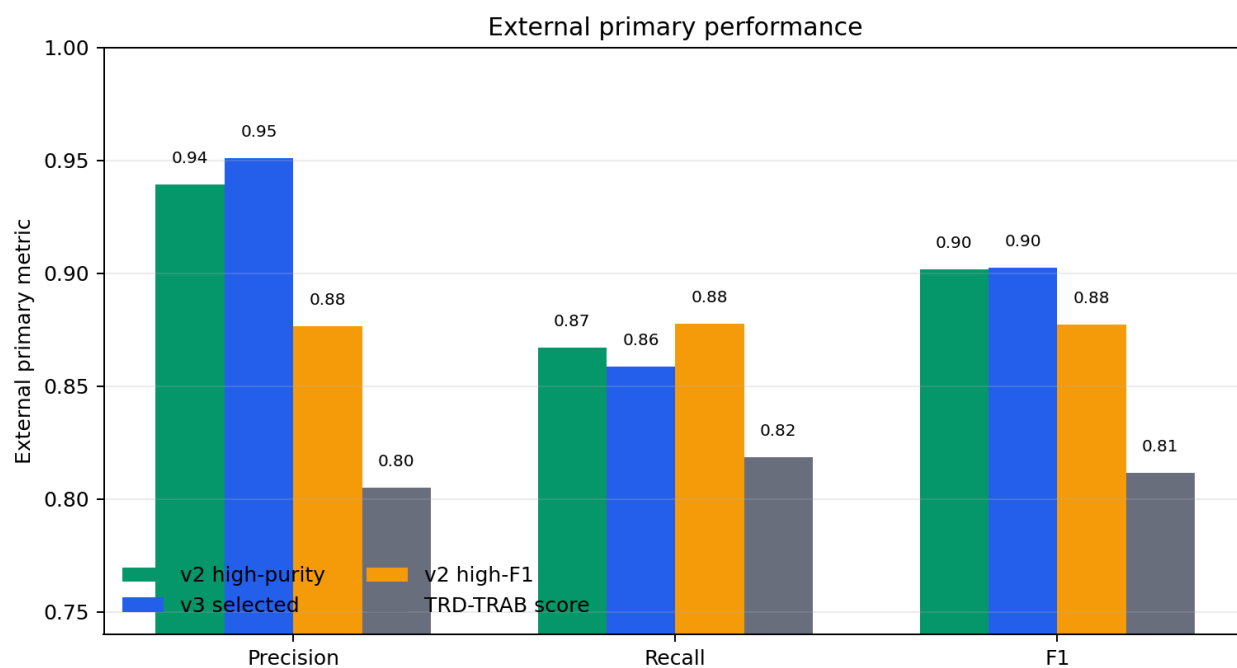
3. How the candidate was selected

The initial internal winner was `v3_round03_hist_gradient_target_recall80_fp05`, but it reached only 72.70% full-atlas primary-gold recall. The final selected round reached 80.56% recall and 4.06% estimated FP/prediction, passing the 80.5% recall-with-margin and 5.0% estimated-FP targets.

External test results were not used for model selection. The best external-F1 v3 row was `v3_round08_hist_gradient_fixed_0p64` with F1 0.9087, but that was evaluated only after the selection decision.



Full-atlas recall versus estimated FP fraction. The selected point passes the target box.



External primary metrics after selection.

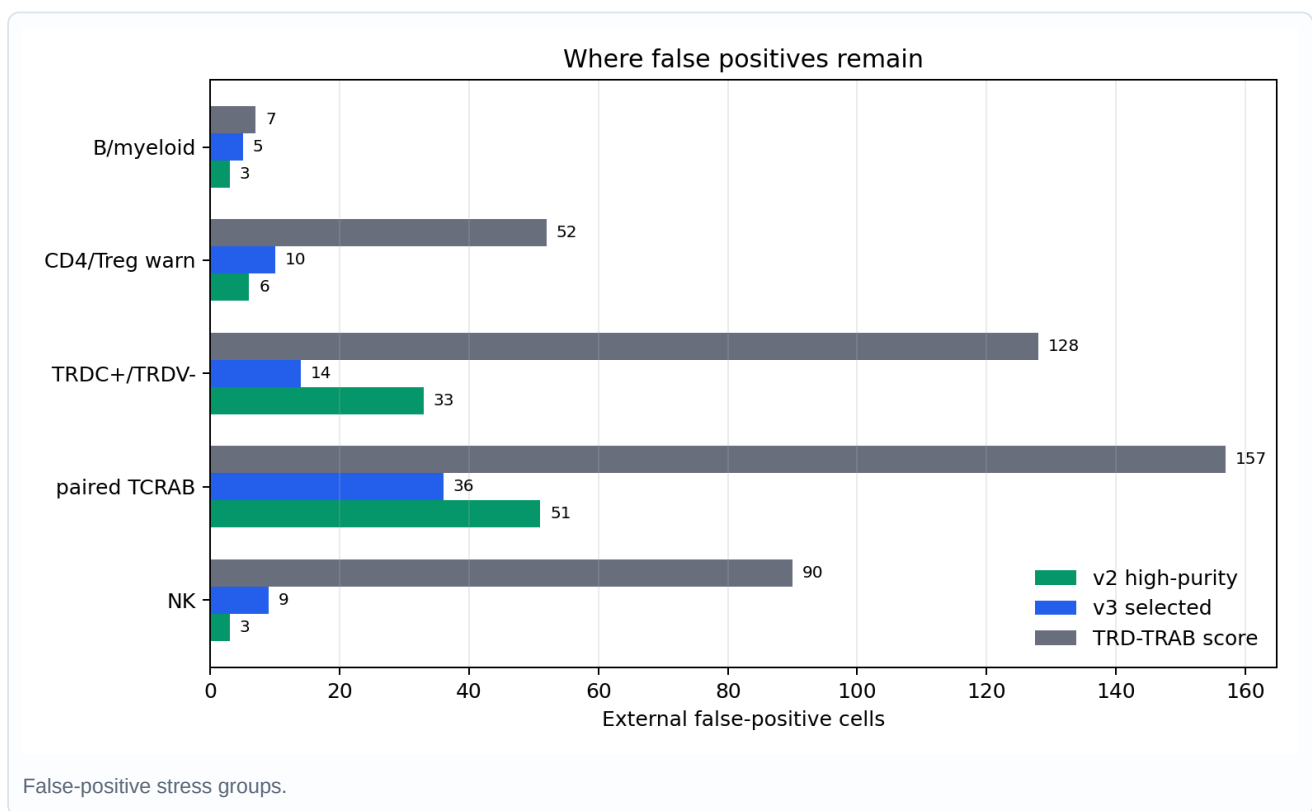
Step conclusion: round14 was selected because it is the first practical operating point satisfying the atlas-level target, not because it looked best on the external test.

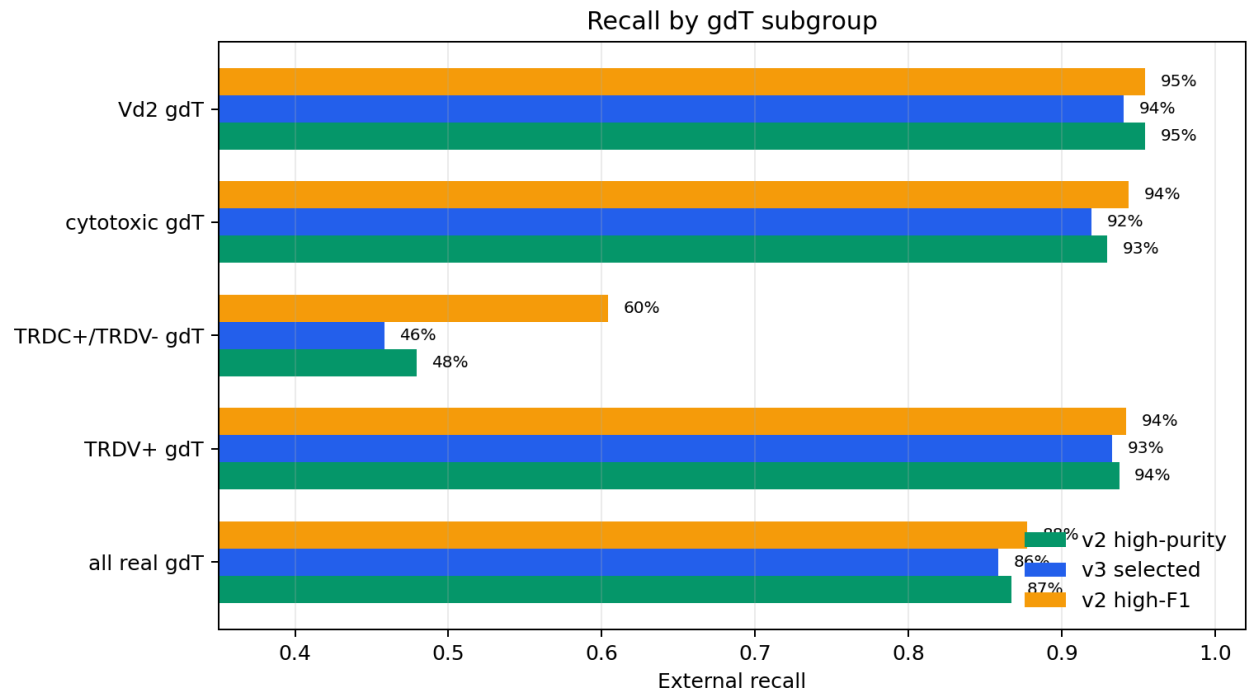
4. External test result

Metric	v3 selected	v2 high-purity	Interpretation
External precision	95.1%	93.9%	v3 higher
External recall	85.9%	86.7%	v3 slightly lower
External F1	0.9024	0.9018	near-tie
External FP / predictions	4.9%	6.1%	v3 lower
External NK FP	9	3	v3 worse

v3 reduced total external false positives from 53 to 42. It also reduced paired-TCRAB false positives from 51 to 36 and TRDC+/TRDV- false positives from 33 to 14.

The problem is NK specificity: NK false positives increased from 3 in v2 high-purity to 9 in selected v3.





Recall stress groups.

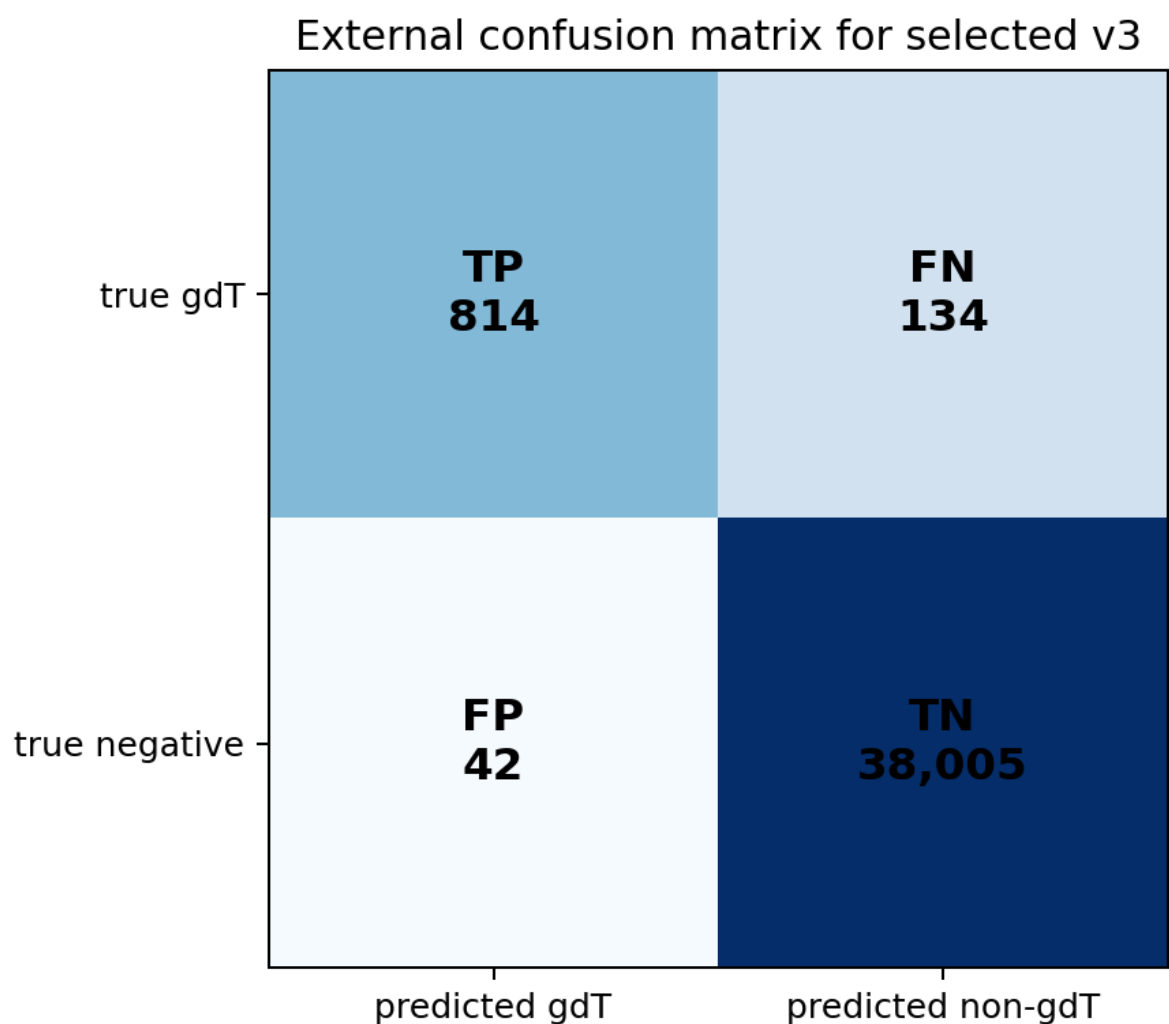
Step conclusion: v3 is competitive overall, but it does not convincingly fix the NK false-positive failure mode.

5. Full-atlas application

Full-atlas metric	v3 selected	v2 high-purity
Predicted putative gdT	251,356	359,857
Primary-gold recall	80.56%	80.56%
Primary-gold precision	97.52%	97.61%
Estimated total abT FP	10,212	9,680
Estimated FP / predictions	4.06%	2.69%
Predicted paired TCRA-B	5,580	5,184
Predicted TRDC+/TRDV-	17,120	130,846

The full-atlas recall is measured on primary gold labels only: $46,544 / (46,544 + 11,232) = 80.56\%$. This is not a claim of absolute biological recall for every unlabeled gdT cell.

The estimated abT FP count uses observed paired-TCRA-B false-positive behavior in TCR-seq sources and extrapolates it to datasets without TCR-seq. It is the best current estimate, but it is still an estimate.



External confusion matrix for selected v3.

Promotion gate checklist

NK FP lower than v2 HP	FAIL
TRDC+/TRDV- burden lower	PASS
F1 not worse by >0.01	PASS
cytotoxic recall preserved	PASS
paired TCRA B FP not higher	PASS

Promotion gate checklist.

Step conclusion: the candidate meets the full-atlas numeric target, but the direct external NK regression prevents promotion.

6. Why this should not be promoted yet

Gate	Status	Evidence
NK false positives lower than v2 high-purity	FAIL	v3 9 vs v2 high-purity 3
TRDC+/TRDV- false-positive burden lower	PASS	v3 14 vs v2 high-purity 33
External F1 not worse by more than 0.01	PASS	v3 0.9024 vs v2 high-purity 0.9018
Cytotoxic gdT recall preserved	PASS	v3 91.9% vs v2 high-purity 93.0%
Paired-TCRA B false positives not higher	PASS	v3 36 vs v2 high-purity 51

The promotion rule was intentionally stricter than overall F1. The previous concern was NK-like contamination, especially around weak TRD evidence. A release model should not improve some aggregate metrics while worsening the named failure mode.

Therefore the correct status is: **package and keep as a v3 candidate**, but do not write it as the promoted `gdTAI_v3.0` default model.

Final conclusion: not promoted because external NK FP is worse than v2 high-purity, while external F1 is essentially unchanged and external recall is slightly lower than v2 high-purity.

7. Detailed artifacts

Large tables are intentionally not embedded in this report. They remain available for audit:

- Model package: `Integrated_dataset/models/gdT_prediction_classifier/gdtai_v3_trdc_nk_guard`
- Full result tables: `Integrated_dataset/tables/gdT_prediction/gdtai_v3_trdc_nk_guard`
- Figures: `gdT_prediction/assets/gdtai_v3_trdc_nk_guard`
- Markdown report:
`Integrated_dataset/logs/gdT_prediction/gdtai_v3_trdc_nk_guard/gdtai_v3_trdc_nk_guard_report.md`
- PDF report: `gdT_prediction/gdtai_v3_trdc_nk_guard_report.pdf`